# 11 THE ELECTROMAGNETIC SPECTRUM

The electromagnetic spectrum is a family of waves, varying in wavelength and frequency. Although it is continuous, it is helpful to consider smaller groups of waves within the spectrum. These groups have distinct properties. As we will see in this chapter, understanding the different properties allows us to use these waves in many situations including cooking and communication.



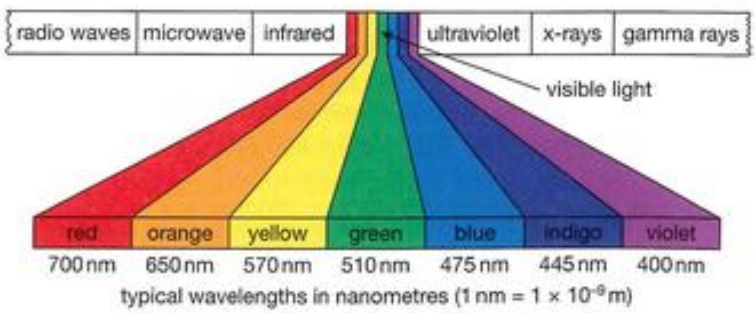
▶ Figure 11.1 When we shine white light through a prism it splits up forming a band of colours. This band is one small part of the electromagnetic spectrum.

## LEARNING OBJECTIVES

- Know that light is part of a continuous electromagnetic spectrum that includes radio, microwave, infrared, visible, ultraviolet, x-ray and gamma ray radiations and that all these waves travel at the same speed in free space
- Know the order of the electromagnetic spectrum in terms of decreasing wavelength and increasing frequency, including the colours of the visible spectrum
- Explain some of the uses of electromagnetic radiations, including:
  - radio waves: broadcasting and communications
  - microwaves: cooking and satellite transmissions
  - infrared: heaters and night vision equipment
  - visible light: optical fibres and photography
  - ultraviolet: fluorescent lamps
  - x-rays: observing the internal structure of objects and materials, including for medical applications
  - gamma rays: sterilising food and medical equipment
- Explain the detrimental effects of excessive exposure of the human body to electromagnetic waves, including:
  - microwaves: internal heating of body tissue
  - infrared: skin burns
  - ultraviolet: damage to surface cells and blindness
  - gamma rays: cancer, mutation

  and describe simple protective measures against the risks

## THE ELECTROMAGNETIC SPECTRUM




▲ Figure 11.2 The complete electromagnetic spectrum

The **electromagnetic spectrum** (EM spectrum) is a continuous spectrum of waves, which includes the visible spectrum. At one end of the spectrum the waves have a very long wavelength and low frequency, while at the other end the waves have a very short wavelength and high frequency. All the waves have the following properties:

1. They all transfer energy.
2. They are all transverse waves.
3. They all travel at 300 000 000 m/s, the speed of light in a **vacuum** (free space).
4. They can all be reflected and refracted.

Remember that the wave equations we met in the previous chapter can be applied to any member of the electromagnetic spectrum.

**EXAMPLE 1**

Yellow light has a wavelength of $5.7 \times 10^{-7}$ m. What is the frequency and period of yellow light waves?

$$v = f \times \lambda$$

$$\text{So } f = \frac{v}{\lambda}$$

$$= \frac{3 \times 10^8 \text{ m/s}}{5.7 \times 10^{-7} \text{ m}}$$

$$= 5.26 \times 10^{14} \text{ Hz}$$

$$T = \frac{1}{f}$$

$$= \frac{1}{5.26 \times 10^{14} \text{ Hz}}$$

$$= 1.9 \times 10^{-15} \text{ s}$$

The table below shows the different groups of waves in order, and gives some of their uses.

| | Typical frequency/Hz | Typical wavelength/m | Sources | Detectors | Uses |
|---|---|---|---|---|---|
| Radio waves | $10^5$–$10^{10}$ | $10^3$–$10^{-2}$ | radio transmitters, TV transmitters | radio and TV aerials | long-, medium- and short-wave radio, TV (UHF) |
| Microwaves | $10^{10}$–$10^{11}$ | $10^{-2}$–$10^{-3}$ | microwave transmitters and ovens | microwave receivers | mobile phone and satellite communication, cooking |
| Infrared (IR) | $10^{11}$–$10^{14}$ | $10^{-3}$–$10^{-6}$ | hot objects | skin, blackened thermometer, special photographic film | infrared cookers and heaters, TV and stereo remote controls, night vision |
| Visible light | $10^{14}$–$10^{15}$ | $10^{-6}$–$10^{-7}$ | luminous objects | the eye, photographic film, light-dependent resistors | seeing, communication (optical fibres), photography |
| Ultraviolet (UV) | $10^{15}$–$10^{16}$ | $10^{-7}$–$10^{-8}$ | UV lamps and the Sun | skin, photographic film and some fluorescent chemicals | fluorescent tubes and UV tanning lamps |
| X-rays | $10^{16}$–$10^{18}$ | $10^{-8}$–$10^{-10}$ | x-ray tubes | photographic film | x-radiography to observe the internal structure of objects, including human bodies |
| Gamma rays | $10^{18}$–$10^{21}$ | $10^{-10}$–$10^{-14}$ | radioactive materials | Geiger–Müller tube | sterilising equipment and food, radiotherapy |

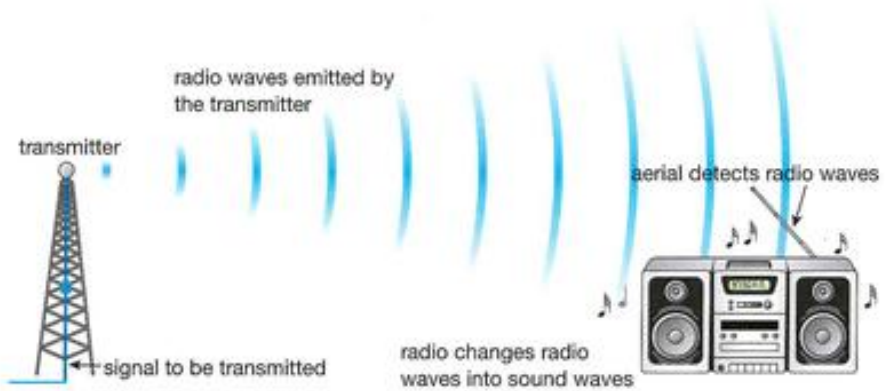
**HINT**

To remember the order of the waves in the electromagnetic spectrum try using 'Graham's Xylophone Uses Very Interesting Musical Rhythms'.

You do not need to remember the values of frequency and wavelength given in the table but you do need to know the order of the groups and which has the highest frequency or longest wavelengths. Most importantly, you need to realise that it is these differences in wavelength and frequency that give the groups their different properties – for example, gamma rays have the shortest wavelengths and highest frequencies, and carry the most energy.

## RADIO WAVES

Radio waves have the longest wavelengths in the electromagnetic spectrum. They are used mainly for communication.

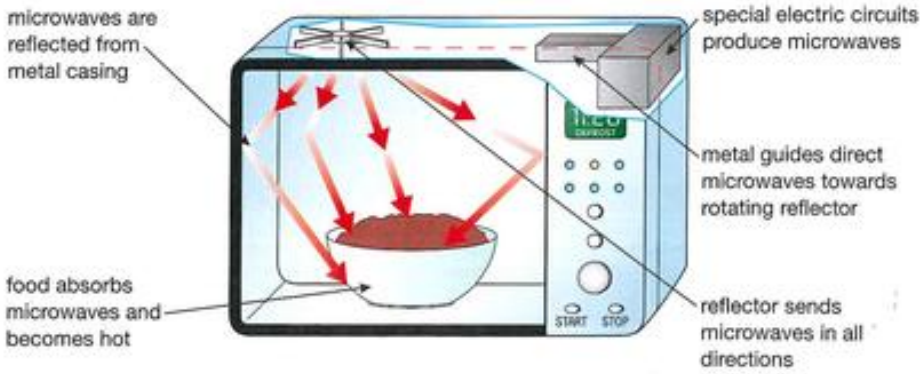



▲ Figure 11.3 Radio waves are emitted by a transmitter and detected by an aerial.

Radio waves are given out (**emitted**) by a transmitter. As they arrive at an aerial, they are detected and the information they carry can be received. Televisions and FM radios use radio waves with the shorter wavelengths to carry their signals.

## MICROWAVES

Microwaves are used for communications, radar and cooking foods. Radar uses radio waves to find the position of things.




▲ Figure 11.4 Food cooks quickly in a microwave oven because water molecules in the food absorb the microwaves.

Food placed in a microwave oven cooks more quickly than in a normal oven. This is because water **molecules** in the food absorb the microwaves and become very hot. The food therefore cooks throughout rather than just from the outside.

Microwave ovens have metal screens that reflect microwaves and keep them inside the oven. This is necessary because if microwaves can cook food, they

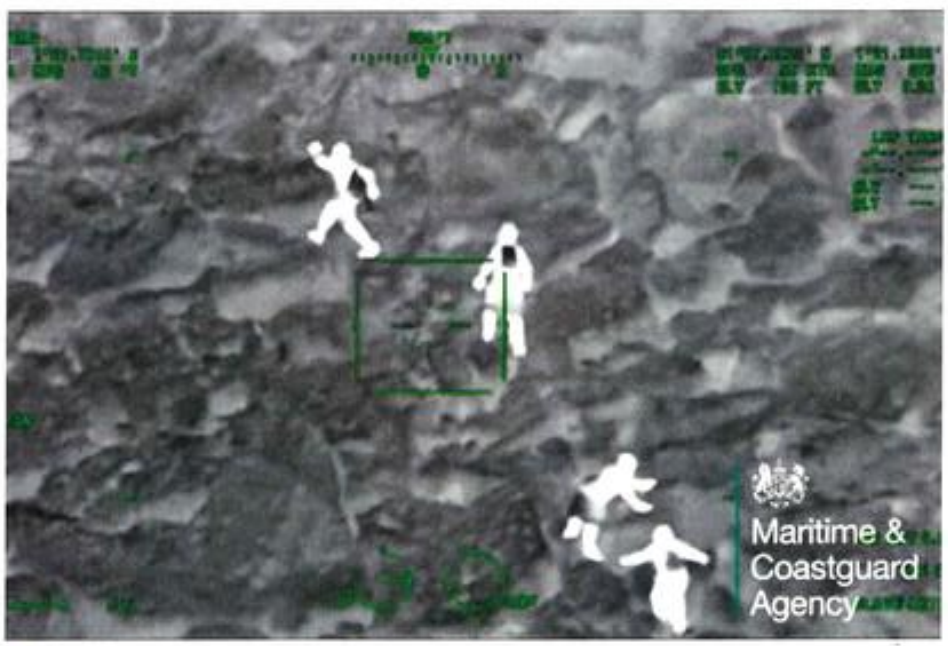
can also heat human body tissue! The microwaves used by mobile phones transmit much less energy than those used in a microwave oven, so they do not cook your brain when you use the phone. However evidence is growing that supports the idea that overuse of mobile phones may eventually harm the brain.

Microwaves are used in communications. The waves pass easily through the Earth's atmosphere and so are used to carry signals to orbiting satellites. From here, the signals are passed on to their destination or to other orbiting satellites. Messages sent to and from mobile phones are also carried by microwaves. The fact that we are able to use mobile phones almost anywhere in the home and at work confirms that microwaves can pass through glass, brick and so on.

## INFRARED

All objects, including your body, emit infrared (IR) radiation. The hotter an object is, the more energy it will emit as infrared. Energy is transferred by infrared radiation to bread in a toaster or food under a grill. Electric fires also transfer heat energy by infrared.

Special cameras designed to detect infrared waves can be used to create images even when there is no visible light. These cameras have many uses, including searching for people trapped in collapsed buildings, searching for criminals and checking for heat loss from buildings.




▲ Figure 11.5 It is not possible to see these people trapped at the bottom of a cliff using normal visible light. By using infrared detectors they can be found easily and rescued.



▲ Figure 11.6 Signals are carried from this remote control to a TV by infrared waves.

Infrared radiation is also used in remote controls for televisions, DVD players and stereo systems. It is very convenient for this purpose because the waves are not harmful. They have a low **penetrating power** and will therefore operate only over small distances, so they are unlikely to interfere with other signals or waves.

The human body can be harmed by too much infrared radiation, which can cause skin burns.

## VISIBLE LIGHT





▲ Figure 11.7 Using visible light

This is the part of the electromagnetic spectrum that is visible to the human eye. We use it to see. Visible light from lasers is used to read compact discs and barcodes. It can also be sent along optical fibres, so it can be used for communication or for looking into hard-to-reach places such as inside the body of a patient (see page 121). Visible light can be detected by the sensors in digital cameras, and used to take still photographs or videos. Information stored on DVDs is also read using visible light.

### EXTENSION WORK

When talking about light and colour, we often refer to the seven colours in the visible spectrum. These colours are red, orange, yellow, green, blue, indigo and violet; red light has the longest wavelength and lowest frequency. If you look back at Figure 11.1, you may only be able to make out six colours – most people have difficulty separating indigo and violet (two types of purple). Sir Isaac Newton (1642–1727) discovered that 'white' light can be split up into different colours. He believed that the number seven had magical significance, and so he decided there were seven colours in the spectrum!

## ULTRAVIOLET LIGHT



▲ Figure 11.8 UV light can cause sunburn so we need to protect our skin.

Part of the light emitted by the Sun is ultraviolet (UV) light. UV radiation is harmful to human eyes and can damage the skin.

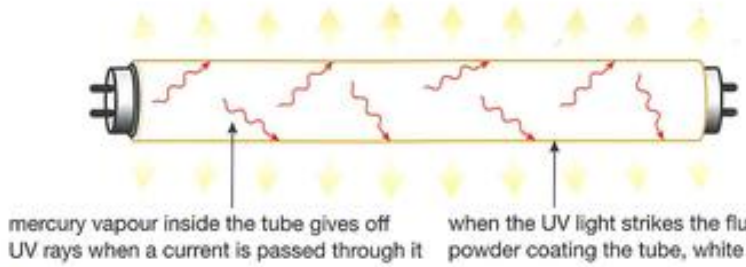
UV light causes the skin to tan, but overexposure (too much) will lead to sunburn and blistering. Ultraviolet radiation can also cause skin cancer and blindness. Protective goggles or glasses and skin creams can block the UV rays and will reduce the harmful effects of this radiation.

The **ozone layer** in the Earth's atmosphere absorbs large quantities of the Sun's UV radiation. There is real concern at present that the amount of ozone in the atmosphere is decreasing due to pollution. This may lead to increased numbers of skin cancers in the future.

Some chemicals glow (shine), or fluoresce, when under UV light. This property of UV light is used in security marker pens. The special ink is invisible in normal light but becomes visible in UV light.




▲ Figure 11.9 This red code is only visible under UV light.

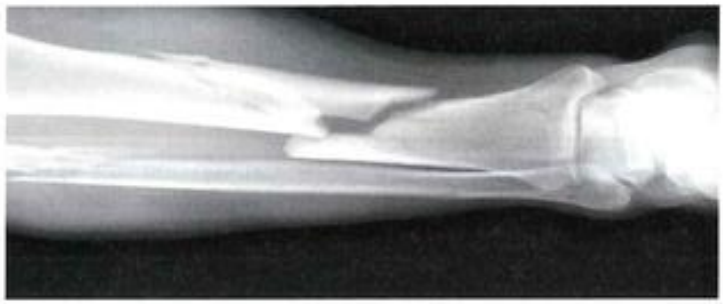



▲ Figure 11.10 Fluorescent tubes glow when UV light hits the fluorescent coating in the tube.

Fluorescent tubes glow (shine) because the UV light they produce strikes a special coating (covering) on the inside of the tube, which then emits visible light.

## X-RAYS

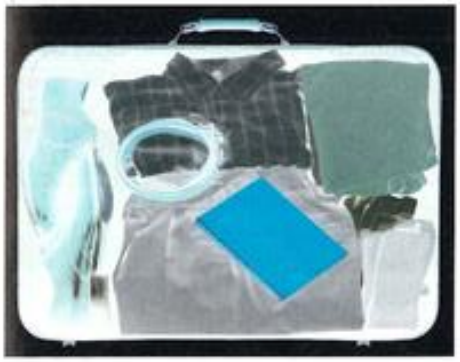
X-rays pass easily through soft body tissue but cannot pass through bones. As a result, radiographs or x-ray pictures can be taken to check a patient's bones.



▲ Figure 11.11 X-ray of a broken leg

Working with x-rays can cause cancer. Radiographers, who take x-rays, are at risk and have to stand behind lead screens or wear protective clothing.

X-rays are also used in industry to check the internal structures of objects – for example, to look for cracks and faults in buildings or machinery – and at airports as part of the security checking procedure.



▲ Figure 11.12 X-rays were used to see what was in this suitcase.

## GAMMA RAYS

Gamma rays, like x-rays, are highly penetrating rays and can cause damage to living cells. The damage can cause mutations (negative changes), which can lead to cancer. They are used to **sterilise** medical instruments, to kill microorganisms so that food will keep for longer and to treat cancer using radiotherapy. Gamma rays can both cause and cure cancer. Large doses of gamma rays targeted directly at the cancerous growth can be used to kill the cancer cells completely.

Like x-rays the use of lead screens, boxes and aprons can prevent the damage caused by gamma rays (overexposure).




▲ Figure 11.13 The gamma rays are aimed carefully so that they cross at the exact location of the cancerous cells.

## CHAPTER QUESTIONS

More questions on using waves can be found at the end of Unit 3 on page 130.

SKILLS CRITICAL THINKING 5

1 a Name four wave properties that are common to all members of the electromagnetic spectrum.

b Name three types of wave that can be used for communicating.

c Name two types of wave that can be used for cooking.

d Name one type of wave that is used to treat cancer.

e Name one type of wave that might be used to 'see' people in the dark.

d Name one type of wave that is used for radar.

SKILLS REASONING 7

2 Explain why:

a microwave ovens cook food much more quickly than normal ovens

b x-rays are used to check for broken bones

c it is important not to damage the ozone layer around the Earth

d food stays fresher for longer after it has been exposed to gamma radiation.

3 a Explain one way in which you could prevent overexposure (damage) by the following waves:

i x-rays

ii ultraviolet waves.

SKILLS CRITICAL THINKING

b Select one of the above waves and then describe one consequence of overexposure.

SKILLS INTERPRETATION 6

4 Copy and complete the table below for four more different wave groups within the electromagnetic spectrum.

| Type of radiation | Possible harm | Precautions |
|---|---|---|
| x-rays | cancer | lead screening |
| | | |
| | | |
| | | |
| | | |